

**Analog & Digital
Communication (EE3003)**

Date: November 04, 2024

Course Instructor(s)

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Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 45

Total Questions: 3

Roll No

Section

Student

Do not write below this line

1. Attempt all questions and remember to solve parts of the same question together.
2. Show all the steps with the help of diagrams and equations.

CLO # 03: Demonstrate the concept of analog amplitude modulation and demodulation techniques

[15 marks]

Q1:

- (a) Describe the principle of operation of a Switching Modulator clearly illustrating (through time and frequency domain representation) the conversion of a message signal, $m(t)$ to a modulated signal of the form $m(t) \cos \omega_c t$. [10]
[Block diagram of the modulator is not required in this part; the concept of periodic on-off switching has to be utilized].
- (b) Draw complete circuit diagram of a "Ring Modulator" and prove that it achieves the switching operation. [5]

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[15 marks]

Q2:

- (a) Define ^{synchronous} coherent demodulation and draw its general block diagram. [4]
- (b) Describe the main difference between envelope detector and rectifier detector as used in conventional AM. [4]
- (c) We want to detect the message signal, $m(t)$ from a conventional AM signal, $[A + m(t)] \cos \omega_c t$, (regardless of the value of A) using coherent detector in part (a) with slight modification. Draw the block diagram and write the equations for each block. (Remember to achieve exactly $m(t)$ at the output). [7]

CLO #04: Demonstrate the concept of analog frequency modulation and demodulation techniques

Q3: An angle-modulated signal with carrier frequency, $\omega_c = 2\pi \times 10^6$ rad/s is described by, [15 marks]

$$\varphi_{EM}(t) = 10 \cos(\omega_c t + 0.1 \sin 2000\pi t)$$

- (a) Find the power of modulated signal. [2]
(b) Find frequency deviation, Δf . [6]
(c) Find phase deviation, $\Delta\varphi$. [2]
(d) Estimate the bandwidth of $\varphi_{EM}(t)$. [5]

mod) $\{A + m(t)\} \cos^2 \omega_c t$

LPF $\left[\frac{1}{2} x(t) + \frac{1}{4} \cos 2\omega_c t x(t) \right]$

$A + m(t) \rightarrow$ DC blocker $\rightarrow m(t)$